

216301003

SEMESTER EXAMINATION-2022
CLASS -B. TECH.
SUBJECT- DIGITAL SYSTEM DESIGN
PAPER CODE: BET-C306

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any five questions in about 150 words each. Each question carries six marks. (5 X 6 = 30 Marks)

Question-1: Realize the following logic operations using only NAND gates: AND, OR, NOT, Ex-OR.

Question-2: Design a BCD adder using IC 7483. — (3)

Question-3: Reduce the following Boolean expression using Boolean laws.
I. $AB + A'B + AB' + (AB)'$
II. $A + B[AC + (B + C')D]$

Question-4: Design a combinational circuit with three inputs and one output. The output is 1 when the binary input is less than 3. The output is 0 otherwise. — (5)

Question-5: Draw JK flip flop and also write the truth table.

Question-6: A combinational circuit is defined by the following three Boolean

$$F1 = x'y'z' + xz$$

$$F2 = xy'z' + x'y$$

$$F3 = x'y'z + xy$$

Design the circuit with a decoder and external gates.

Question-7: Define prime implicants (PI) and Essential prime implicants (EPI) with an example.

Question-8: Minimize the following function using k map and realize using NAND gate.
 $\sum m(0,2,3,6,7) + d(8,10,11,15)$ (5)

Question-9: Draw the schematic and explain the operation of CMOS NOR gate.

Question-10: Implement the following function using 8:1 MUX
 $F(A, B, C, D) = \sum (1, 3, 4, 11, 12, 13, 14, 15)$

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any FOUR questions in detail. Each question carries 10 marks.
 (4 X 10 = 40 Marks)

Question-11: Simplify following function Boolean expression using Tabular Method:

$$F(A, B, C, D) = \sum m(0, 1, 3, 4, 5, 7, 10, 13, 14, 15)$$

Question-12: Perform following operations:

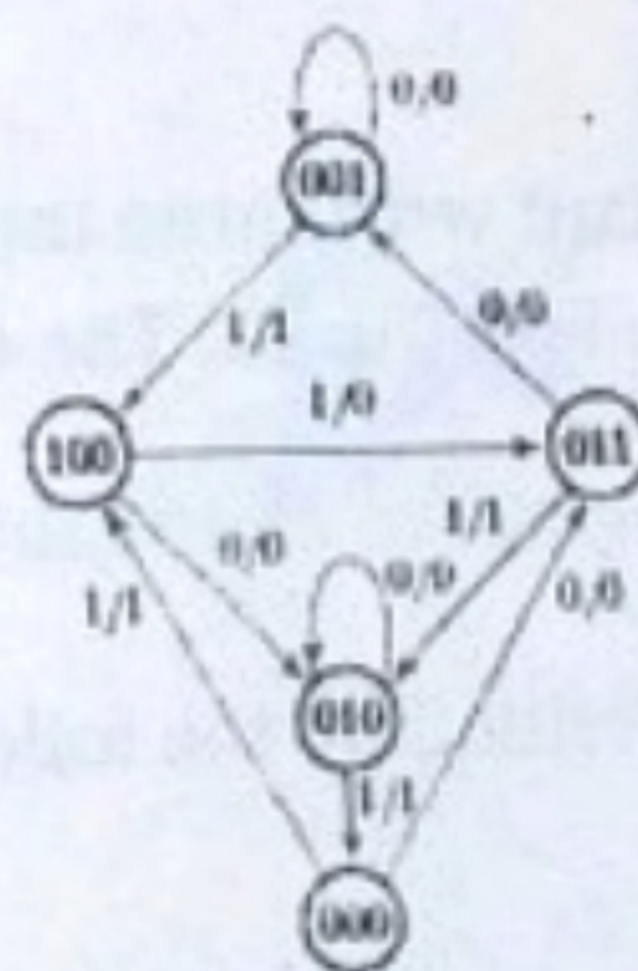
- i. add 42 to -55 using 8 bit 2's complement arithmetic.
- ii. Convert $(564)_{10}$ to hexadecimal.
- iii. Convert $(124)_8$ to Gary code.
- iv. add 297.20 to 483.50 in BCD code.

Question-13: (a) Realize T FF using SR FF.

(b) Construct 16:1 MUX using two 8:1 MUX and one 2:1 MUX

Question-14: Design MOD 10 Asynchronous counter using T flip flop.

Question-15: A sequential circuit has three flip-flops A, B, C; one input x; and one output y. The state diagram is shown in Fig. The circuit is to be designed by treating the unused states as don't-care conditions. Analyze the circuit obtained from the use of JK Flip flop in design.



Question-16: Explain the operation of universal shift register with neat and clean block diagram.

Question-17: Design a synchronous counter using D-FF with the following repeated binary sequence: 0, 1, 2, 4, 6. Check whether counter is self-starting or not. If not, then make the solution.

Question-18: Explain the operation of 2 input TTL NAND gate with required diagram. What are the advantages and disadvantages of totem pole?

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